

Bumblebee Colony Multimodal Imaging Architecture

A Technical Integration Guide for High-Resolution, Thermal, and Depth Mapping Using Dual Raspberry Pi 5 Hardware.

1. System Layout & Hardware Partitioning

Running an Arducam 64MP camera at full resolution (2.5 FPS), a PureThermal 3 / FLIR Lepton 3.5 core, and an Intel RealSense depth camera simultaneously on a single Raspberry Pi 5 will lead to severe system memory bottlenecks and I/O thrashing. Splitting the cameras across a **dual Raspberry Pi 5 (8GB)** configuration balances execution workloads and guarantees deterministic frame captures.

Raspberry Pi A: High-Resolution Imaging Logger

- **Primary Load:** Arducam 64MP Camera connected via the MIPI CSI-2 port.
- **Compute Profile:** High CPU utilization. Capturing a raw or uncompressed 64MP image requires approximately 90MB to 120MB of buffer RAM per frame. Debayering, encoding, and committing these massive arrays to storage at 2.5 FPS introduces significant pipeline overhead. This board should be entirely isolated from other peripheral streams.

Raspberry Pi B: Multimodal Sensor Aggregator

- **Primary Load:** Intel RealSense Depth Camera + PureThermal 3 / Lepton 3.5 Core connected via dedicated USB 3.0 ports.
- **Compute Profile:** Balanced CPU utilization. The Intel RealSense features an onboard Vision Processor D4 ASIC that calculates complex spatial depth maps natively before shipping data over the wire. The Lepton 3.5 sensor pushes a lightweight 160x120 radiometric thermal array. Pi B operates with significant processing headroom, protecting against buffer drops.

2. Sensor Performance & Selection Specifications

Feature	Arducam 64MP Camera	Intel RealSense D435i	FLIR Lepton 3.5
Data Type	High-Res Visible Spectrum	Active IR Depth Stream	Long-Wave IR Radiometric
Resolution	9152 × 6944 (64 MP)	Up to 1280 × 720	160 × 120
Bus Interface	MIPI CSI-2 (Camera Port)	USB 3.1 Gen 1 Type-C	USB UVC (via PT3)
Min Range	Lens Dependent (Macro)	0.11 meters	Lens Dependent

3. Cross-Platform Synchronization Strategies

Because data pipelines are split physically between two operating systems, you must implement a robust strategy to correlate frames during post-processing analysis.

Method A: Precision Network Time Protocol (PTP)

Link both Raspberry Pi 5 boards directly using a standard Gigabit Ethernet patch cable. Configure one Pi as a local PTP master server. This aligns their internal system clocks to sub-millisecond precision. Capture scripts running on both machines append the system epoch timestamp down to the millisecond directly into every recorded file name (e.g., frame_1719124410123.raw). Your analysis pipeline can then easily align the 64MP frame, the Lepton thermal data, and the RealSense depth array via matching filenames.

Method B: Physical Hardware Level Triggering (Genlock)

For absolute microsecond-level synchronization of fast-moving bumblebee interactions, utilize a hardware-driven trigger signal:

1. **Master Pulse:** Use a designated GPIO pin on Raspberry Pi B to output a 3.3V high logic pulse exactly 2.5 times per second.
2. **64MP Interrupt:** Route this physical trigger wire into a GPIO input pin on Raspberry Pi A. Configure the Arducam capture script to wait for a hardware interrupt, forcing an immediate image capture on the voltage spike.
3. **RealSense Genlock:** Splice the same trigger wire into a 3.3V to 1.8V level shifter (as RealSense logic requires 1.8V pulses). Connect the shifted output to **Pin 5 (Sync)** on the RealSense external sync header under its rubber housing cap. Program the camera via the librealsense API into *Inter-Cam Sync Mode 4*. The depth sensor will hold exposure and only snap frames precisely on the external hardware pulse.